AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM271
Machine Learning and Deep Learning

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 4/Dec. /2025

Number of Exam Pages: 2 (including this cover sheet)	**Time Allowed: 120 Minutes**

Instructions:

- Total Marks: 30
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part – A (5 Marks)

1. In _____ learning, the system is incapable of learning incrementally.
   a. Online
   b. batch
   c. SVM
   d. Decision tree
2. SVM is capable of _____
   a. classification
   b. regression
   c. outlier detection
   d. All the above
3. _____ is the technique where the predictor aggregates the prediction of multiple predictors.
   a. Decision Tree
   b. SVM
   c. Ensemble learner
   d. All the above
4. Which one of the following is Visualization and dimensionality reduction technique?
   a. PCA
   b. Kernel PCA
   c. LLE
   d. All the above
5. _____ is used only for linear classification.
   a. Linear Regression
   b. Decision Tree
   c. SVM
   d. Decision Tree

Part – B (15 Marks)

Provide brief answers to the following questions.

1) Explain any three commonly used clustering techniques? (3 Marks)
2) Explain any three different types of techniques in Machine Learning. (3 Marks)
3) Compare the Batch Gradient Descent vs stochastics Gradient Descent techniques (6 Marks)
4) What is the role of kernal function in deep learning techniques? (3 Marks)

Part – C (10 Marks)

NOTE: This question does not need any programming code.

A bank wants to develop a machine learning model to predict whether a customer will default on a loan based on features such as income, credit score, number of dependents, loan amount, and previous repayment history. As a data scientist, explain how you would use a Support Vector Machine (SVM) model to build this classifier. Your answer should include the main steps involved from data loading to model evaluation.

Instructions:

- Provide a solution for building a recommendation system.
- Present your solution in a stepwise manner.
- Ensure that your response consists of a minimum of four steps.

End of questions

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